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Vegetation-driven and passive monitoring of urban particulate matter: Comparative field assessment in Fukuoka, Japan

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1. Abstract & Scope

Urban particulate matter (PM) poses a severe health hazard. This study compared vegetation-based PM capture with an environmentally friendly, Ferm-type passive diffusive sampler under identical roadside conditions in Fukuoka, Japan. Foliar PM accumulation was quantified for three roadside plant functional types: Elaeagnus pungens (shrub), Dioscorea japonica (climber), and Cirsium vulgare (herb). The passive sampler accumulated significantly higher PM mass and showed enrichment in smaller particle fractions than the foliage. This was due to the sampler's physical stability, which avoided the particle shedding and rain wash-off experienced by leaves. The study concluded that vegetation reflects real-world, dynamic deposition and wash-off cycles, while passive samplers provide stable, long-term integrated measurements.

2. Methodology & Experimental Parameters

Roadside sampling in Fukuoka. Foliar wash-off vs. starch-based passive sampler capture. Scanning Electron Microscopy (SEM) mapping of leaf topography and gravimetric analysis.

3. Core Findings & Data Synthesis

Foliar capture was driven by micromorphology (grooves, margins, hairs) rather than leaf area. The passive sampler provided higher stability and smaller fraction enrichment.

4. Strategic Relevance to JU and Kolkata Planning

Proves that plants are dynamic sensors reflecting deposition/removal cycles, and suggests pairing them with physical passive samplers to calibrate urban baselines.